NMC 2013 O-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn. 1. The mathematician Augustus lived his entire life during the 1800's. In the last year of his life he stated: At one time, I was x years old in the year x^2. In which year was Augustus born?

Qn. 2. In the subtraction below a, b and c are digits and a is less than b. What is the value of c?

$$ba - ab = c6$$

Qn. 3. There are five sheds located on a farm. The distance between shed A and B is 2 km, between B and C is 1.5 km, between C and D is 8 km and between D and E is 3.5 km. Finally shed A is 1 km from shed E. How far is shed B from shed E?

Qn. 4. If $ab = 12$ and $(a + 1)(b + 1) = 23$, what is the value of $(a + 2)(b + 2) + 4$?

Qn. 5. Suppose that $\frac{97}{19} = w + \frac{1}{x+\frac{1}{y}}$ where w, x, y are all integers. Find the value of $w + x + y$.

Qn. 6. To make one can of orange paint, 3 parts of red paint are mixed with 2 parts of yellow paint and to make one can of green paint, 2 parts of blue paint are mixed with 1 part of yellow paint. If equal volumes of green and orange paints are mixed, What is the proportion of yellow paint in the mixture?

Qn. 7. A restaurant can accommodate at most 400 people, which includes the guests and the waiters. Each waiter can serve a maximum of 12 guests. What is the greatest number of guests that can be served ?

Qn. 8. In the garden of Kristin, there are two plants. One is 44 cm high, and it grows 3 cm in each 2 years. The other is 80 cm high, and it grows 5 cm in each 6 years. In how many years will the two plants be equally high?

Qn.9. James has a three- digit code for a padlock. He has forgotten the code but he knows that if you divide the first digit by the second digit and then square the result, you get the third digit. How many three digit codes have this property?. Show your working.

Qn.10. Each of the following five statements is either true or false. How many more statements are true given that statement (4) is true.

(1) Statements (3) and (4) are both true.

(2) Statements (4) and (5) are not both false.

(3) Statement (1) is true.

(4) Statement (3) is false.

(5) Statements (1) and (3) are both false.

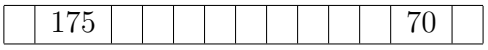
SECTION B

Qn.11. Find the total number of three-digit numbers that are divisible by 9 and contain no even digits.

Qn.12. On each face of a cube there is a different number from 1 to 6. If one of the faces is at the bottom of the cube, then the sum of the numbers on all the faces except the top and bottom is 13. If another face is at the bottom, then the sum of the numbers on all the faces except the top and bottom is 12. What is the number written on the opposite side of the face labelled 1?

Qn.13. Abel, Berna, Carol, Dan and Eva play a game of cards. If Berna gives away three cards, she will remain with the same number of cards as Eva. If Berna is instead given three more cards, she will have as many cards as Dan. Abel and Berna together have 10 cards more than the total cards for Dan and Eva. If Berna has two cards more than Carol and the total number of cards is 133, how many cards does Berna have?

Qn.14. The diagram below is to be completed so that each box contains a whole number. The total of the numbers in the thirteen boxes is 2005 and the sum of the numbers in any three consecutive boxes is always the same. In how many different ways is it possible to complete the diagram in this way?



Qn.15. Let a and b be the lengths of the two shorter sides of the right angled triangle. C_1 and C_2 are the circles circumscribing and inscribing the triangle respectively. If the longest side D of the triangle is the diameter of C_1 and d is the diameter of C_2, express $D + d$ in terms of a and b.

NMC 2013 O-LEVEL PAPER 1 SOLUTIONS

SECTION A : 5 marks each

1. $y = \frac{2}{x}, y = \frac{24}{z}, z = \frac{3}{x}$. This means $\frac{24}{z} = \frac{2}{x} \Longrightarrow z = 12x$. Also $12x = \frac{3}{x} \Longrightarrow x^2 = \frac{1}{4}$.
   $x = \pm\frac{1}{2}$. This then implies that $y = 4$ and $z = 3 \times 2 = 6$.
   Therefore $x + y + z = \frac{1}{2} + 4 + 6 = 10\frac{1}{2}$.

2. Since AB is prime, B is either 1 or 3. Also $A + C \leq 11$ so $E = 1$. It then follows that $B = 3$. Now, since $D \leq 6$, $B + D < 10$ and $A + C = 10$. We deduce that A and C must be 4 and 6 in some order but since 63 is not prime, $AB = 43 \Longrightarrow A = 4$.
   Therefore $A + B = 4 + 3 = 7$

3. Since a and b are the roots of the equation, then $x - a$ and $x - b$ are factors of the quadratic equation. Therefore

$$\begin{aligned} x^2 + ax + b &= (x - a)(x - b) \\ x^2 + ax + b &= x^2 + (a + b)x + ab \\ \text{equating coefficients } \ b &= ab \ \text{ and } \ a = -a - b. \\ \text{since } \ b \neq 0 &\Longrightarrow a = 1 \text{and so } \ b = -2 \end{aligned}$$

   ALTERNATIVELY
   we substitute a and b in the quadratic equation to get $2a^2 + b = 0$ and $b^2 + ab + b = 0$ implying $b = -2a^2$ and $b = -a - 1 \Longrightarrow -2a^2 - a - 1 = 0$. Solving this quadratic equation gives WWait abit

4. $\sqrt[3]{\frac{1\times2\times4+2\times4\times8+3\times6\times12+4\times8\times16+\cdots}{1\times3\times9+2\times6\times18+3\times9\times27+4\times12\times36+\cdots}} = \sqrt[3]{\frac{(1\times2\times4)(1+8+27+64+\cdots)}{(1\times3\times9)(1+8+27+64+\cdots)}} = \sqrt[3]{\frac{8}{27}} = \frac{2}{3}$.

5. $f(x + 1, y) = f(x, y) + y + 1$, $f(x, 0) = x$ and $f(x, y) = f(y, x)$ for all x and y
   $f(12, 0) = 12 = f(0, 12)$. Then,
   $f(1, 12) = f(0, 12) + 12 + 1 = f(0, 12) + 13 = 12 + 13 = 25$.
   $f(2, 12) = f(1, 12) + 12 + 1 = f(1, 12) + 13 = 25 + 13 = 38$.
   $f(3, 12) = f(2, 12) + 12 + 1 = f(2, 12) + 13 = 38 + 13 = 51$.
   $f(4, 12) = f(3, 12) + 12 + 1 = f(3, 12) + 13 = 51 + 13 = 64$.
   Hence $f(4, 12) = f(12, 4) = 64$.

6.

7. Consider the following table of statements of 4 statements depending on who has the key.

Ann has the key	F	F	T	F
Ben has the key	T	F	F	T
Cal has the key	F	T	F	T
Don has the key	T	T	T	F

Hence Don has the key.

8. Let $N = 19700019y$. Clearly, y cannot be even or 5. If y is a multiple of 3, then N is divisible by 3 which can be checked by summing its digits and obtain a multiple of 3. As 197 is an apparent divisor of N if $y = 7$, then the only possibility is when $y = 1$.

9. We observe that $m.[m] = 17 \Longrightarrow 4 < m < 5$ so that $[m] = 4$. Therefore $m = \frac{17}{4}$.
Similarly, $n.[n] = 17 \Longrightarrow 3 < n < 4$ so that $[n] = 3$. Thus $n = \frac{11}{3}$

10. Suppose there are b boys and g girls in the family. Then, each boy has $(b-1)$ brothers and g sisters. Hence $b - 1 = g \cdots\cdots (1)$
Each girl has b brothers and $g - 1$ sisters. This implies $b = 2(g - 1)$. we can substitute for b in the equation (1) to get $2g - 2 - 1 = g$
$\Longrightarrow g = 3$ and so $b = 4$.
The total number of children in the family is $b + g = 4 + 3 = 7$.

SECTION B : 10 **marks each**

11. Suppose there are x girls and y boys in the class and that a cake costs p coins and a sandwich costs q coins. we know that

$$\begin{aligned} xp + yq &= yp + xq - 1 \\ (xp - yp) + (yq - xq) &= -1 \\ p(x - y) - q(x - y) &= -1 \\ (x - y)(p - q) &= -1. \end{aligned}$$

Since both factors are integers, one must be $+1$ and the other -1. We also know that $x > y$. This implies $x - y = 1$. Therefore there is only one additional girl.

12. Out of the ten sums $32, 36, 37, \cdots, 48, 51$ the largest one is $d + e$ and the second largest one is $c + e$. Therefore $d + e = 51$ and $c + e = 48$.
Furthermore, $a + b$ is the smallest sum and $a + c$ the second smallest, so $a + b = 32$ and $a + c = 36$.
The third sum could be either $a + d$ or $b + c$.
However, we know that $a + d = (a + c) + (d + e) - (c + e) = 36 + 51 - 48 = 39$.
Apparently, $a + d$ is not the third smallest sum, so $b + c = 37$.
Combining the things we have got so far yields
$2e = 2(c + e) - (a + c) - (b + c) + (a + b) = 2(48) - 36 - 37 + 32 = 55$
This implies that $2e = 55 \Longrightarrow e = \frac{55}{2}$.

13. (a) Let the first bricklayer lay x bricks per hour and the second bricklayer y bricks per hour. The total number of bricks in the wall $= 9x = 10y$. Working together, they lay $x + y - 10$ bricks per hour. The total number of bricks in the wall $= 5(x + y - 10)$.
Now, $5(x + y - 10) = 9x \Longrightarrow 5y - 4x = 50 \cdots\cdots\cdots\cdots\cdots$(i).
Also $5(x + y - 10) = 10y \Longrightarrow -5y + 5x = 50 \cdots\cdots\cdots\cdots$ (ii).
Solving equations (i) and (ii) simultaneously gives $x = 100$ and $y = 90$. The total number of bricks in the wall is 900 bricks.

    (b) Suppose the time period is t days during which it rains on x mornings and y afternnons. It never rains on both morning and afternoon of the same day.
The number of rain days $= x + y = 11 \cdots\cdots\cdots\cdots\cdots\cdots$ (i)
The number of clear mornings $= t - x = 9 \cdots\cdots\cdots\cdots\cdots$ (ii)
The number of clear afternoons $= t - y = 12 \cdots\cdots\cdots\cdots$ (iii)
Adding equations (i), (ii) and (iii) gives $2t = 32 \Longrightarrow t = 16$.
The total number of rain free days is $t - 11 = 5$ days.

14. It is given that $x, y > 0$ hence $0 \neq x + y = x^2 - y^2 = (x + y)(x - y)$. Divide through $(x + y)$ to get $x - y = 1 \Longrightarrow x = y + 1$.
We also know that $xy = x+y$ hence $0 = xy-y-x = x(y-1)-y = (y+1)(y-1)-y = y^2-y-1$

    so y is positive and $y^2 = y + 1$ and solving this quadratic equation gives $b = \frac{1+\sqrt{5}}{2}$.

15. Sketch

    Drop a perpendicular from P to side AB and let R be the point of intersection. Let $\overline{AR} = x$ and $\overline{BR} = y$. Similary, drop a perpendicular from P to side AD and let S be the point of intersection .
Let $AS = u, DS = v \Longrightarrow y^2 + u^2 = \overline{PB}^2 = 4^2 = 16$.

    Also, $u^2 + x^2 = \overline{PA}^2 = 8^2 = 64$.

    $x^2 + v^2 = \overline{PD}^2 = 7^2 = 49$.
Hence $\overline{PC}^2 = y^2 + v^2 = (y^2 + u^2) + (x^2 + v^2) + (u^2 + x^2) = 16 + 49 - 64 = 1$. Thus, $\overline{PC} = 1$.